

# Requirements

## Cohort 1 Group 7

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### Introduction

The requirements were gathered from the product brief and a customer meeting. First, the product brief was read and core deliverables were collated. A meeting was then organised with the customer where the deliverables were discussed and refined with more understanding. The customer also added additional requirements that were not present in the product brief and a single statement of need was gathered. After the meeting the user requirements were imputed into a table with an unique ID and how important the feature is; a table was used because it makes it clear and concise for anyone involved in the project to understand and use the data. From the user requirements, another two tables were created for system requirements: functional and non-functional, each with the ID from the user requirements table, another unique ID and a description that can be measured. These are also presented in tables as it

allows the developers to code and test each requirement quickly and easily ensuring all essential deliverables are met.

### Statement of user and system requirements

**SSON** - A single-player desktop game that doesn't last more than 5 minutes and has the player escaping a situation at university

### User requirements table

|                     |                                                                                      |        |
|---------------------|--------------------------------------------------------------------------------------|--------|
|                     |                                                                                      |        |
| UR_SINGLEPLAYER     | The game shall be singleplayer                                                       | Shall  |
| UR_TIME             | The game shall not take longer than 5 minutes                                        | Shall  |
| UR_DEAN             | The game shall have a dean who chases the player                                     | Shall  |
| UR_NEGATIVE_EVENTS  | The game shall have 5 negative visible events that disadvantage the player           | Shall  |
| UR_POSITIVE_EVENTS  | The game shall have 5 positive visible events that advantage the player              | Shall  |
| UR_HIDDEN_EVENTS    | The game shall have 3 hidden events that either advantage or disadvantage the player | Shall  |
| UR_ESCAPE           | The game shall have a way for the player to escape                                   | Shall  |
| UR_PAUSE            | The game shall have a way to pause the game                                          | Shall  |
| UR_SCORING_SYSTEM   | The game shall have a scoring system based on time and events                        | Shall  |
| UR_OPERATING_SYSTEM | The game shall be playable on all operating systems                                  | Shall  |
| UR_TUTORIAL         | The game should have an introductory scene to give the player a tutorial             | Should |
| UP_MAP              | The game should consist of a single large map                                        | Should |
| UR_STYLE            | The game should be played from a top down perspective                                | Should |

|                 |                                                               |        |
|-----------------|---------------------------------------------------------------|--------|
| UR_POWER_UP     | The game should give the player abilities from events         | Should |
| UR_STAMINA      | The game may have a stamina bar that limits the players speed | May    |
| UR_DIFFICULTIES | The game may have different difficulties                      | May    |

### Functional requirements table

|                 |                                                                         |                   |
|-----------------|-------------------------------------------------------------------------|-------------------|
|                 |                                                                         |                   |
| FR_SINGLEPLAYER | The game will only support singleplayer                                 | UR_SINGLEPLAYER   |
| FR_FIVE_MINUTE  | The game will not allow the player to play for more than 5 minutes      | UR_TIME           |
| FR_EXIT_FIVE    | The game may end before 5 minutes if the players finds an exit          | UR_TIME           |
| FR_DEAN_CHASE   | The game will have a dean that chases the player                        | UR_DEAN           |
| FR_DEAN_CATCH   | The dean will not end the game                                          | UR_DEAN           |
| FR_DEAN_LIMIT   | The dean will limit the players movement                                | UR_DEAN           |
| FR_NEGATIVE     | The game will show the player at least 5 choices that disadvantage them | UR_NEGATIVE_EVENT |
| FR_POSITIVE     | The game will show the player at least 5 choices that advantage them    | TS                |
| FR_HIDDEN       | The game will have at least 3 events that have a hidden trigger         | UR_POSITIVE_EVENT |
|                 |                                                                         | S                 |
|                 |                                                                         | UR_HIDDEN_EVENTS  |
| FR_ESCAPE       | The game has a way for the player to win by escaping the map            | UR_ESCAPE         |

|                     |                                                                                         |                     |
|---------------------|-----------------------------------------------------------------------------------------|---------------------|
| FR_PAUSE            | The game will include a pause function to stop gameplay and timer                       | UR_PAUSE            |
| FR_SCORING_SYSTEM   | The game will calculate a score based on elapsed time and triggered events              | UR_SCORING_SYSTEM   |
| FR_OS_SUPPORT       | The game will be playable across major operating systems                                | UR_OPERATING_SYSTEM |
| FR_TUTORIAL_SCENE   | The game should include an introductory tutorial scene to teach controls and objectives | UR_TUTORIAL         |
| FR_SINGLE_MAP       | The game should consist of one large continuous map                                     | UP_MAP              |
| FR_TOP_DOWN_VIEW    | The game should use a top-down camera perspective                                       | UR_STYLE            |
| FR_POWER_UPS        | The game will provide the player with special abilities through events                  | UR_POWER_UP         |
| FR_STAMINA_BAR      | The player may have a stamina bar that limits movement speed                            | UR_STAMINA          |
| FR_DIFFICULTY_MODES | The game may include multiple difficulty modes                                          | UR_DIFFICULTIES     |

### Non functional requirements table

|                   |                                                                             |                  |                                                                  |
|-------------------|-----------------------------------------------------------------------------|------------------|------------------------------------------------------------------|
|                   |                                                                             |                  |                                                                  |
| NFR_TIME_LIMIT    | The game must ensure that a complete play session does not exceed 5 minutes | UR_TIME          | Each game session automatically ends after 5 minutes of gameplay |
| NFR_HIDDEN_EVENTS | The game must display a hidden event when it is triggered                   | UR_HIDDEN_EVENTS | The hidden events will show within 1 second of the trigger       |

|                        |                                                                                                      |                     |                                                                                       |
|------------------------|------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------|
| NFR_OS_COMPATIBILITY   | The game shall run on all major operating systems (Windows, macOS, Linux)                            | UR_OPERATING_SYSTEM | The game launches and runs without errors on all three OS environments during testing |
| NFR_OS_PERFORMANCE     | The game shall maintain consistent performance and frame rate across all supported operating systems | UR_OPERATING_SYSTEM | Average frame rate $\geq 30$ FPS on standard hardware across all tested OS versions   |
| NFR_VISUAL_STYLE       | The game should use a top-down perspective to maintain visual clarity and consistency                | UR_STYLE            | Gameplay camera remains fixed above the player perpendicular to the ground            |
| NFR_USER_INTERFACE     | The user interface should be intuitive and minimalistic to support top-down gameplay                 | UR_STYLE            | 90% of test users can navigate the UI and start a game without external guidance      |
| NFR_STABILITY          | The game shall not crash or freeze during normal operation                                           | UR_OPERATING_SYSTEM | 0 critical crashes over 10 consecutive games                                          |
| NFR_LOAD_TIME          | The game should load completely within 5 seconds on a standard desktop computer                      | UR_OPERATING_SYSTEM | Average load time $\leq 5$ seconds measured over 5 startup tests on standard hardware |
| NFR_USABILITY_TUTORIAL | The tutorial should be clear and concise, enabling a new player to learn controls quickly            | UR_TUTORIAL         | 80% of first-time players complete the tutorial and start gameplay within 2 minutes   |